Climate Change, Stock Productivity, and Fishing Effort in Chile's Multi-Species Small Pelagic Fishery

DRAFT – PLEASE DO NOT CITE

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abril 17, 2026

Abstract

We study how climate change will affect stock productivity and fishing effort in Chile's multi-species small pelagic fishery, composed of anchoveta (*Engraulis ringens*), jack mackerel (*Trachurus murphyi*), and sardine (*Strangomera bentincki*). We estimate a three-equation Seemingly Unrelated Regression (SUR) model for stock growth as a function of biomass, sea surface temperature, and chlorophyll-a, and a negative binomial model for annual fishing trips as a function of prices, quotas, weather, and regulatory closures. Using CMIP6 projections from the IPSL-CM6A-LR model under SSP2-4.5 and SSP5-8.5 scenarios, we apply the delta method to project changes in environmental conditions at mid-century (2041–2060) and end-of-century (2081–2100). We decompose the projected climate impact on fishing effort into a direct weather channel (wind affecting operational days) and an indirect biomass channel (temperature and chlorophyll affecting stock productivity and harvest allocations). Results reveal a stark asymmetry: the artisanal fleet benefits from warming-driven sardine expansion (effort increases of 20–250%), while the industrial fleet faces declining jack mackerel productivity (effort declines of approximately 22%). The indirect biomass channel dominates, with the direct weather effect contributing less than 5% of the total response. These findings highlight heterogeneous climate vulnerability across fleet segments and have implications for quota allocation policy in multi-species fisheries.

1 Introduction

The distribution and abundance of marine resources are changing in response to environmental conditions such as global ocean warming (**?**). Climate change will shift species distribution in the

future, leading to reduced species availability in some areas and increased availability in others (**?**). The literature that studies fishers' responses to either changes in fish availability or policies that restrict access to fisheries (e.g., **?**; **?**) has identified that they can adopt the following adaptive strategies: (i) reduce or reallocate fishing effort, either to another species or to another location (**?**); (ii) continue following the same strategy; or (iii) exit the fishery and find alternative employment (**?**). Among these strategies, reallocating effort to alternative species has been identified as a potentially effective response to climate change (**?**). Diversification of target species has also been linked to reduced income variability (e.g., **?**; **?**) and greater resilience to both climate shocks (**?**; **?**) and interannual oceanographic variability (**?**; **?**).

This emphasis on diversification aligns with broader evidence from food-producing sectors. As Cruz (**?**) highlights, climate variability substantially affects agriculture and fisheries, where income depends heavily on environmental fluctuations (e.g., temperature, rainfall) and market forces (e.g., input costs) (**?**; **?**). With variability expected to reduce productivity, income risk is likely to rise (**?**; **?**). Diversification—whether within a sector (e.g., switching crops or species) or across sectors—is often promoted as an adaptive strategy (**?**). However, these strategies can be costly for resource-dependent communities with limited capital and skills (**?**; **?**), and the role of switching costs in shaping diversification remains poorly understood.

In the fisheries context, switching between species requires not only the skills but also the appropriate gear and permits (**?**; **?**). Even if these conditions are met, diversification may still be constrained by port infrastructure, markets, and regulations (**?**; **?**; **?**). Therefore, deciding which adaptation strategy to adopt is not straightforward and depends on multiple factors. Moreover, fishers may respond differently to similar circumstances depending on their goals, skills, and preferences (**?**; **?**; **?**).

In this research, we aim to answer how fishing decisions, aggregate catch levels, and the price of marine resources will be affected under different climatic scenarios in the multi-species small pelagic fishery (SPF) in Chile, composed of anchoveta (*Engraulis ringens*), jack mackerel (*Trachurus murphyi*), sardine (*Strangomera bentincki*), among others. The SPF is the most important in terms of catches in the country, accounting for almost 94% of the total Chilean catch in 2019 (**?**). Through this research, we aim to gain a deeper understanding of how Chilean fishers and fishing communities will adapt to climate change. According to Cheung et al. (**?**), the Chilean Exclusive Economic Zone (EEZ) is projected to experience one of the largest losses in maximum catch potential due to climate change. However, the Southeast Pacific remains one of the least studied regions regarding the impacts of climate change on fisheries (**?**).

To address this question, we estimate a multi-species bioeconomic model inspired by Kasperski (**?**). This model captures the biological and economic interrelations among species to study how climate variability affects stock productivity and fishing effort. We estimate a three-equation SUR system for stock growth dynamics and a negative binomial model for annual fishing trips, and combine these with CMIP6 climate projections to project changes in fishing activity under different

emission scenarios. We decompose the climate impact into a direct weather channel (wind affecting operational feasibility) and an indirect biomass channel (temperature and chlorophyll affecting stock growth and harvest allocations).

Under a changing climate, studying the effect of climatic variability on fishers' harvest decisions and landings is relevant for understanding fishing communities' adaptive capacities and strategies in response to climate change, thereby enabling the design of potential mitigation measures in response to these changes by policymakers. Countries have different institutions, cultures, and norms, leading to differing responses based on the study's location. For this reason, conducting this research based on the Chilean fishing industry is necessary to develop local policies that aim to reduce climate change impacts on fisheries. While there is some literature on the effect of climate change on Chilean fisheries, I am unaware of local-level studies that consider a multiple-species framework and the interrelationship between the local market and fishing decisions seen under a variable climate context.[1]

Because predator–prey links couple these species, reductions in anchoveta or sardine availability may reflect not only environmental drivers but also changes in predation pressure from jack mackerel (**?**; **?**). Thus, even fishers who do not target jack mackerel can face induced changes in catch rates and revenues through ecosystem feedbacks. We hypothesize that if the availability of a main target species decreases, fishers will switch to the closest substitute when expected revenues (net of switching costs) exceed expected costs; otherwise they may reduce effort or exit the fishery. We also expect cross-fleet spillovers: in Chile, jack mackerel is predominantly harvested by the industrial purse-seine fleet, whereas anchoveta and sardine have substantial artisanal and industrial participation. Shocks in one component can propagate across fleets via both biology and markets, in addition to economic linkages (bycatch constraints, shared gear, and market spillovers). This strengthens the case for a multi-species framework that models joint dynamics and substitution rather than single-species responses. Moreover, we expect that this behavior is heterogeneous depending on the geographical area of operation—as it determines the availability of other species (**?**)—and the gear type used.

– ADD the fact that the government define the TAC using sa single species model. This modelling effort allow to move further to a better management of the SPF fishery by using a multispecies approach to obtain optimal TAC

2 SPF in Chile

The small pelagic fishery (SPF) in Chile is of critical importance to the national fisheries sector. In 2019, the SPF represented nearly 94% of total national fish landings (**?**). The fishery is primarily

[1] For the case of Chile, as far as I know, the only article that studies fishers' behavior using discrete choice modeling is Peña-Torres et al. (**?**). This article studies how the El Niño–Southern Oscillation (ENSO) affects fishers' location choices in the jack mackerel fishery.

composed of anchoveta (*Engraulis ringens*), sardine (*Strangomera bentincki*), and jack mackerel (*Trachurus murphyi*). While in the Northern region competition mainly occurs between anchoveta and jack mackerel, in the Central-South region all three species play a major role. This makes the Central-South particularly relevant for the study of species interactions and potential substitution within a multispecies management framework, and it is therefore the focus of this research.

The jack mackerel fishery was initially concentrated in northern Chile, but since the mid-1980s the main fishing grounds have shifted to Central-South Chile, traditionally within 50 nautical miles of the coast (**?**). Historically, species in the SPF have been used primarily for fishmeal and fish oil production (**?**). In fact, about 85% of jack mackerel landings, on a yearly average between 1987 and 2004, were destined for reduction into fishmeal and fish oil (**?**). Today, several key ports serve as hubs for these activities, including San Antonio, Tomé, Talcahuano, San Vicente, Coronel, Lota, and Corral.

"La pesquería de sardina común (Strangomera bentincki) y anchoveta (Engraulis ringens) de la zona centro sur de Chile se caracteriza por ser una pesquería mixta. A pesar de contener especies distintas, éstas conviven y se reproducen en un mismo hábitat. Las artes de pesca utilizadas para capturar estas especies no permiten diferenciarlas." (**?**)

"La pesquería de sardina común y anchoveta ha sido por largos años una de las pesquerías más importante de Chile. La disminución paulatina de los desembarques de jurel (Trachurus murphyi) ha llevado a la sardina común a posicionarse como el principal recurso pelágico extraído, seguido por la anchoveta" (**?**)

2.1 Status of the stocks

Historically, anchoveta in the Central-South was considered collapsed until 2018, shifted to overexploited status in 2019, and has since 2020 been fished within maximum sustainable yield (MSY) limits. Meanwhile, sardine stocks have generally remained within MSY levels, except in 2021 and 2023 when they were classified as overexploited. Jack mackerel was overexploited until 2018 but has since been harvested within MSY limits.

2.2 Quota allocation

The Chilean fishing sector is managed primarily through a Total Allowable Catch (TAC; *Cuota Global*), which is divided between the industrial and artisanal sectors. A small share (2%) is reserved for research, with additional portions allocated to contingency and human consumption. The TAC is subdivided by region and season, and unused quotas may be reassigned during the fishing year.

Anchoveta and sardine are regulated as a mixed-species fishery: although each has its own quota, substitution between them is permitted. A share of industrial quota is also periodically reassigned

to the artisanal sector.

Since 2013, the industrial sector has operated under an individual transferable quota (ITQ) system, known as Transferable Fishing Licenses (*Licencias Transables de Pesca, LTP*). Class A licenses were allocated based on historical catches, while Class B licenses—up to 15% of the industrial fraction—are auctioned, with the first auctions held in 2015. These sealed-bid, first-price auctions aimed to broaden access and limit concentration but have faced challenges such as low participation, difficulties in reflecting economies of scale, and signs of potential coordinated bidding (**?**).

The artisanal TAC operates under a regulated freedom-to-fish regime, allowing registered vessels to fish without individual quotas, except in areas where access is closed or suspended, in which case authorities may implement management measures. The main measure is the Régimen Artesanal de Extracción (RAE), which allocates the regional artisanal TAC by area, vessel size, landing site (caleta), organization, or individually, in agreement with artisanal fisher organizations. To date, area-based and organization-based allocations are the only observed schemes. Area-based allocations allow registered artisanal vessels in a given area to fish as in open access until the assigned quota is exhausted, while organization-based allocations follow the historical rights of members to distribute the organization's quota.

- Sardine: RAE in V, VIII Y X regions? What about other species? Open access in anchovy and jack mackerel (only artisanal TAC matter at country level?)

2.2.1 Chile regionalized fisheries governance framework

Chile has a regionalize fisheres governance framework, where boat register in one region can not fish in another one. For instance, a recent conflict between the Biobío and Ñuble regions has reignited the debate over the spatial governance of the small pelagic fishery (SPF) in south–central Chile. In late August 2025, the Chilean Chamber of Deputies approved a resolution urging the Government to repeal the authorization that allows vessels from Biobío to operate in the coastal waters of Ñuble. The measure, promoted by local authorities and artisanal organizations from Ñuble, aims to protect local fishing grounds and reduce pressure on nearshore ecosystems. However, representatives from Biobío have warned that such restrictions could have severe economic consequences for the region, given its strong dependence on small pelagic landings. This episode highlights the institutional tensions arising from Chile's regionalized fisheries governance framework, where jurisdictional boundaries often conflict with the biological and economic interdependencies of fish stocks.

2.3 Other regulations

2.3.1 Limited entry

Fishery with restricted access to new operators

2.3.2 Biological closures for recruitment

- Jack mackerel is open through all year.
- Sardine and anchovy: In southern-central Chile, December–March (fixed period: January to February).

2.3.3 Biological closures for reproduction

- Jack mackerel is open through all year.
- Sardine and anchovy: In southern-central Chile, July–October (fixed period: August-September).
- Seasonality? Include quarter dummies? Jack mackerel gather in the first 6 month of the year in shoals, great density in EEZ, then migrate outside 200nm ()

2.3.4 Minimum size

- Jack mackerel: 26 cm
- Sardine and anchovy?

2.3.5 Maximum harvest levels

- All species: Maximum catch limit per vessel owner (LMC) for industrial vessels, based on the industrial share of the TAC.

2.4 Switching patterns

ARE THIS SPF TRIPS?????

See Figure **??** for strategy transitions. The year 2019 is used as reference as anchoveta and jack mackerel started to recover.

Table **??** for strategy transitions.

Table **??** for industrial strategy transitions.

Table 1: Comparison of Strategies Before and After – Small-scale vessels

	Before		After	
Strategy	n	%	n	%
Sardine and Anchoveta	420	31.9	376	63.5
Only Sardine	416	31.6	133	22.5
Sardine and Other	193	14.6	8	1.4
Sardine, Anchoveta and Other	139	10.5	21	3.5
Sardine, JackMackerel and Anchoveta	23	1.7	23	3.9
Only Other	60	4.6	2	0.3
Only Anchoveta	21	1.6	16	2.7
Anchoveta and Other	14	1.1	2	0.3
Sardine and JackMackerel	10	0.8	3	0.5
Only JackMackerel	7	0.5	3	0.5
JackMackerel and Other	4	0.3	2	0.3
JackMackerel and Anchoveta	1	0.1	3	0.5
JackMackerel, Anchoveta and Other	4	0.3	0	0.0
Sardine, JackMackerel, Anchoveta, Other	4	0.3	0	0.0
Sardine, JackMackerel and Other	2	0.2	0	0.0

Table 2: Comparison of Strategies Before and After – Industrial vessels

	Before		After	
Strategy	n	%	n	%
Only JackMackerel	46	36.2	28	96.6
Sardine and JackMackerel	22	17.3	1	3.4
Sardine and Anchoveta	14	11.0	0	0.0
JackMackerel and Other	13	10.2	0	0.0
Sardine, JackMackerel and Anchoveta	13	10.2	0	0.0
Only Other	6	4.7	0	0.0
JackMackerel and Anchoveta	3	2.4	0	0.0
Sardine, JackMackerel and Other	3	2.4	0	0.0
Only Sardine	2	1.6	0	0.0
Anchoveta and Other	1	0.8	0	0.0
Only Anchoveta	1	0.8	0	0.0
Sardine and Other	1	0.8	0	0.0
Sardine, Anchoveta and Other	1	0.8	0	0.0
Sardine, JackMackerel, Anchoveta, Other	1	0.8	0	0.0

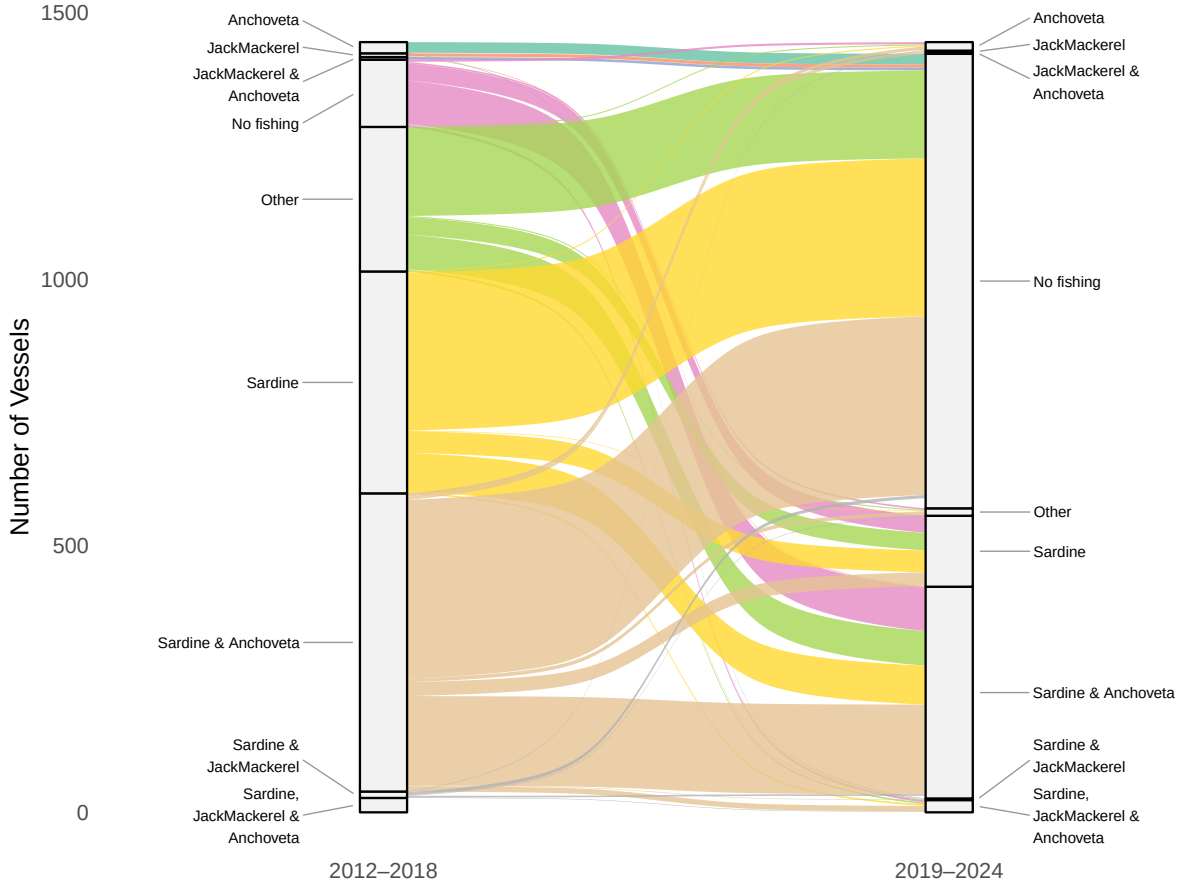

Figure 1: Strategy transitions for small-scale vessels

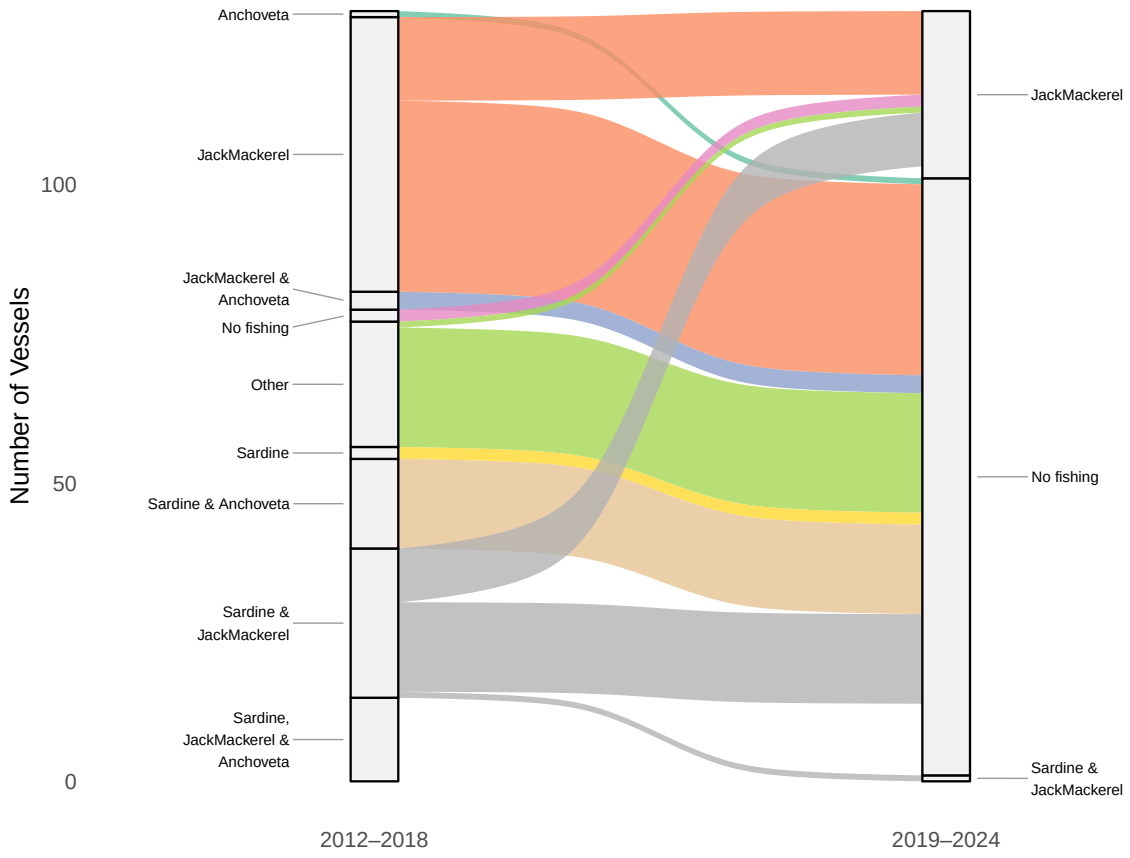

Figure 2: Strategy transitions for industrial vessels

2.5 Fishing seasons

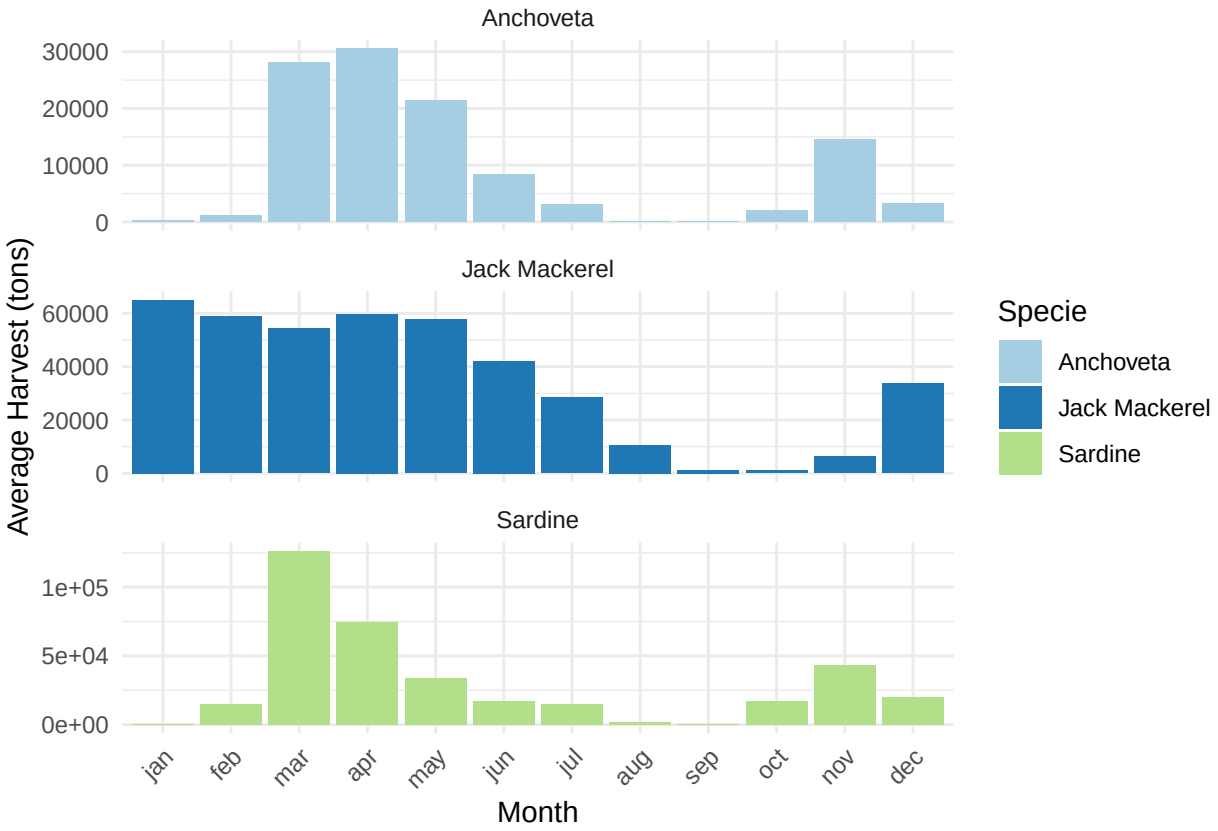

Figure 3: Average monthly landings by species (2012-2024; South-Central Chile)

3 Data and methodology

Our empirical approach has three stages. First, we estimate a multi-species stock dynamics model using a SUR framework to capture how environmental conditions (SST, chlorophyll-a) affect the growth capacity of each species. Second, we estimate a negative binomial model for annual fishing trips as a function of prices, harvest allocations, weather conditions, and regulatory closures, separately for the artisanal and industrial fleets. Third, we combine the estimated models with CMIP6 climate projections to assess how climate change will affect stock productivity and fishing effort through both direct (weather) and indirect (biomass) channels. A companion paper extends this framework to include trip-level cost functions, an inverse demand system, and full numerical optimization of quota paths following Kasperski (**?**).

3.1 Historical data

We use data requested from the Chilean Fisheries Development Institute (Instituto de Fomento Pesquero, IFOP) covering the 2013–2024 period. The dataset includes trip-level microdata, which contain detailed records on vessel identifiers, departure and arrival times, vessel capacity, fleet and gear type, ports of departure and landing, fishery codes, haul timing and location, species composition, retained catch, and trip activity. In addition, we requested annual information on stock abundance and vessel landings by port, county, region, country, and species. Finally, we use ex-vessel prices, reported monthly or annually by port, county, region, country, and species. These prices reflect those paid by processing plants to fishers at the point of first sale and are obtained through IFOP's landing surveys, which do not necessarily cover all market transactions.

For the environmental covariates, we use data from the E.U. Copernicus Marine Service Information, accessed through the Copernicus Marine Toolbox API. Salinity, sea surface temperature, and current speed and direction were obtained from the Global Ocean Physics Reanalysis (GLORYS12V1), which provides data at a 1/12° horizontal resolution with 50 vertical levels (**?**). Wind speed and direction at the surface were obtained from the Global Ocean Hourly Reprocessed Sea Surface Wind and Stress from Scatterometer and Model dataset, available at 0.125° horizontal spatial resolution and hourly frequency (**?**). Chlorophyll-a concentrations were obtained from the Global Ocean Colour dataset, which provides data at ~4 km horizontal resolution (**?**). All environmental data were retrieved daily (hourly in the case of winds) for the 2013–2024 period, covering the Chilean Exclusive Economic Zone (EEZ) between 32°S and 41°S (Figure **??**)

Imputation of missing jack mackerel biomass To address the years in which no acoustic survey was conducted in south-central Chile—or in which insufficient information was available to produce a stock assessment—we imputed missing jack mackerel biomass values. Of the 25 years in the sample period (2000–2024), jack mackerel CS biomass is directly observed in 17 years; the remaining 8 years require imputation. The primary predictor is jack mackerel biomass from the Northern zone, estimated from independent acoustic surveys conducted since 2010 (correlation with observed CS biomass: $r = 0.82$, $N = 7$ overlapping years).

We estimated five alternative GLM specifications with a Gamma distribution and log link, ranging from a parsimonious model with Northern biomass as the sole predictor (2 parameters) to a saturated specification including Northern biomass, its square, the SPRFMO spawning biomass index, and their interaction (5 parameters). All models using Northern biomass are estimated with only 7 observations (years where both CS and Norte are simultaneously observed). The most complex specification exhibited severe overfitting (pseudo-$R^2 = 0.97$, but biologically implausible predictions requiring ad hoc censoring in two years). We therefore selected a parsimonious specification—Model B: Northern biomass and its square (3 parameters)—which achieves an in-sample pseudo-$R^2 = 0.90$ and a leave-one-out cross-validation $R^2 = 0.78$, indicating acceptable

out-of-sample predictive performance. For the single year where Northern biomass is also unavailable (2022), we use linear interpolation between adjacent observed values as a fallback.

The SPRFMO spawning biomass index, which covers the transzonal stock across the EEZ of Chile, Ecuador, and Peru, was also considered as a predictor. However, its correlation with observed CS biomass is low ($r = 0.11$, $N = 17$), reflecting the weak correspondence between regional stock dynamics and local abundance in south-central Chile. The interpolated series, together with a source flag indicating whether each observation is acoustically observed, GLM-predicted, or linearly interpolated, is used in subsequent estimation. To assess sensitivity to the imputation procedure, we report robustness checks using only observed jack mackerel biomass (Appendix **??**).

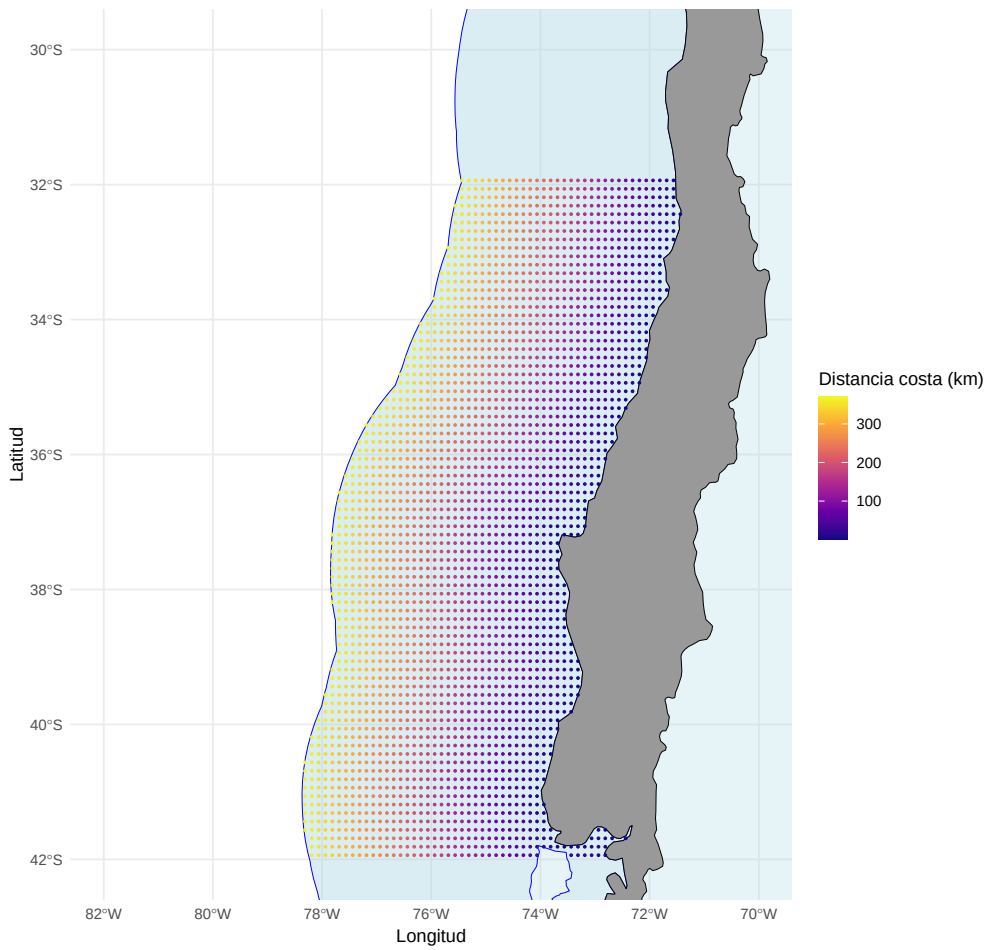

Figure 4: Geographical extent of the study area used for environmental covariates limited to the Chilean Exclusive Economic Zone

3.1.1 To be requested

- Diesel cost.
- Permits by vessels

- Quota prices?
  - Auction market but also secondary market if available
    - If no data, maybe intrapolate prices from other auctions?
  - Captures elements of forward-looking behavior and information (**?**). Reimer et al. (**?**) similarly argue that including a quota price captures forward looking behavior and allows one to simplify the dynamic model to a static one.
- Quota by area/fishing organization for Artisanal, and TAC for industrial with ITQ (by vessel?)

3.2 Future climate data

Climate projections for SST, chlorophyll-a, and near-surface wind speed are obtained from the CMIP6 archive, using the IPSL-CM6A-LR Earth system model. We use monthly outputs for two future scenarios—SSP2-4.5 (moderate) and SSP5-8.5 (high emissions)—and two time windows—mid-century (2041–2060) and end-of-century (2081–2100)—against a 1995–2014 historical baseline. Details of the delta-method projection approach are presented in Section **??**.

3.3 Econometric models

3.3.1 Stock dynamics

To estimate stock dynamics, we use annual data on species-specific biomass (stock abundance) and vessel landings within Chile's Exclusive Economic Zone (EEZ). Following Kasperski (**?**), the baseline model for the interannual growth of each species i is represented by a discrete logistic function that accounts for intra- and inter-species interactions:

$$x_{i,y+1} + h_{iy} = \underbrace{(1+r_i)x_{iy} + \eta_i x_{iy}^2}_{R_i(x_{iy})} + \underbrace{\sum_{j \neq i}^{n-1} a_{ij} x_{iy} x_{jy}}_{I_i(x_y)} \quad i = 1, \dots, n \tag{1}$$

where x_{iy} is the biomass of species i in year y, n is the total number of species, h_{iy} is the annual harvest, r_i is the intrinsic growth rate, η_i is the density-dependent parameter, and α_{ij} captures pairwise species interactions. The system of n growth equations is estimated simultaneously using Seemingly Unrelated Regression (SUR). Following Richter et al. (**?**), we augment (**??**) by including environmental covariates Env_{iy} that affect stock dynamics, along with an error term ε_{iy} representing stochastic recruitment:

$$x_{i,y+1} + h_{iy} = \underbrace{(1+r_i)x_{iy} + \eta_i x_{iy}^2}_{R_i(x_{iy})} + \underbrace{\sum_{j \neq i}^{n-1} a_{ij} x_{iy} x_{jy}}_{I_i(x_y)} + \rho_i Env_{iy} + \varepsilon_{iy} \quad i = 1, \dots, n \tag{2}$$

where ρ_i denotes the environmental response parameters. Each species has its own equation, allowing for species-specific growth, density dependence, and environmental sensitivities.

Environmental conditions were summarized annually using sea surface temperature (SST) and chlorophyll-a concentration (CHL), both averaged over the South-Central Chile region within the EEZ. These variables were selected due to their recognized influence on small pelagic productivity and spatial distribution (**?**). SST serves as a proxy for large-scale oceanographic regimes (warm vs. cold phases) that shape recruitment success along the Humboldt Current System. Anchoveta typically dominates during cold, nutrient-rich phases, whereas sardine tends to increase under warmer regimes (**?**; **?**). CHL reflects interannual variation in primary productivity, which forms the energetic base sustaining small pelagics (see **?**; **?**).

To select the preferred specification, we compared five candidate models by AIC and BIC, varying the inclusion of quadratic environmental terms, cross-species interaction terms, and wind covariates (Table **??**). Wind effects were excluded first, as a joint exclusion test failed to reject their omission. Cross-species interaction terms ($x_{iy}x_{jy}$) and quadratic CHL were subsequently tested but did not improve fit—both AIC and BIC increased when these terms were added. The preferred specification retains own biomass, its square, SST, SST^2, and CHL (linear) for each species, yielding six parameters per equation and approximately 17 residual degrees of freedom with $N = 23$ observations. This parsimonious model is estimated as:

$$x_{i,y+1} + h_{iy} = (1+r_i)x_{iy} + \eta_i x_{iy}^2 + \rho_{i,1} SST_y + \rho_{i,2} SST_y^2 + \rho_{i,3} CHL_y + \varepsilon_{iy} \quad i = 1, \dots, n \tag{3}$$

where all environmental variables are centered at their sample means to reduce collinearity between linear and quadratic terms.

Figure **??** shows that anchoveta, sardine, and jack mackerel exhibit distinct biomass trajectories over time, with no evidence of strong interannual co-movement across species. Biomass levels differ markedly between them, and the figure suggests only limited interrelation alongside clear divergences in year-to-year dynamics. Harvest pressure is also visibly associated with these fluctuations; for instance, jack mackerel experienced an abrupt decline during the late 2000s, consistent with the combined effects of intense fishing pressure and unfavorable environmental conditions.

The SUR framework is particularly suited to this multi-species setting, as it allows for contemporaneous correlation among residuals that naturally arise from shared non-observed environmental shocks, trophic linkages, and imperfectly observed ecological processes. We estimate a three-

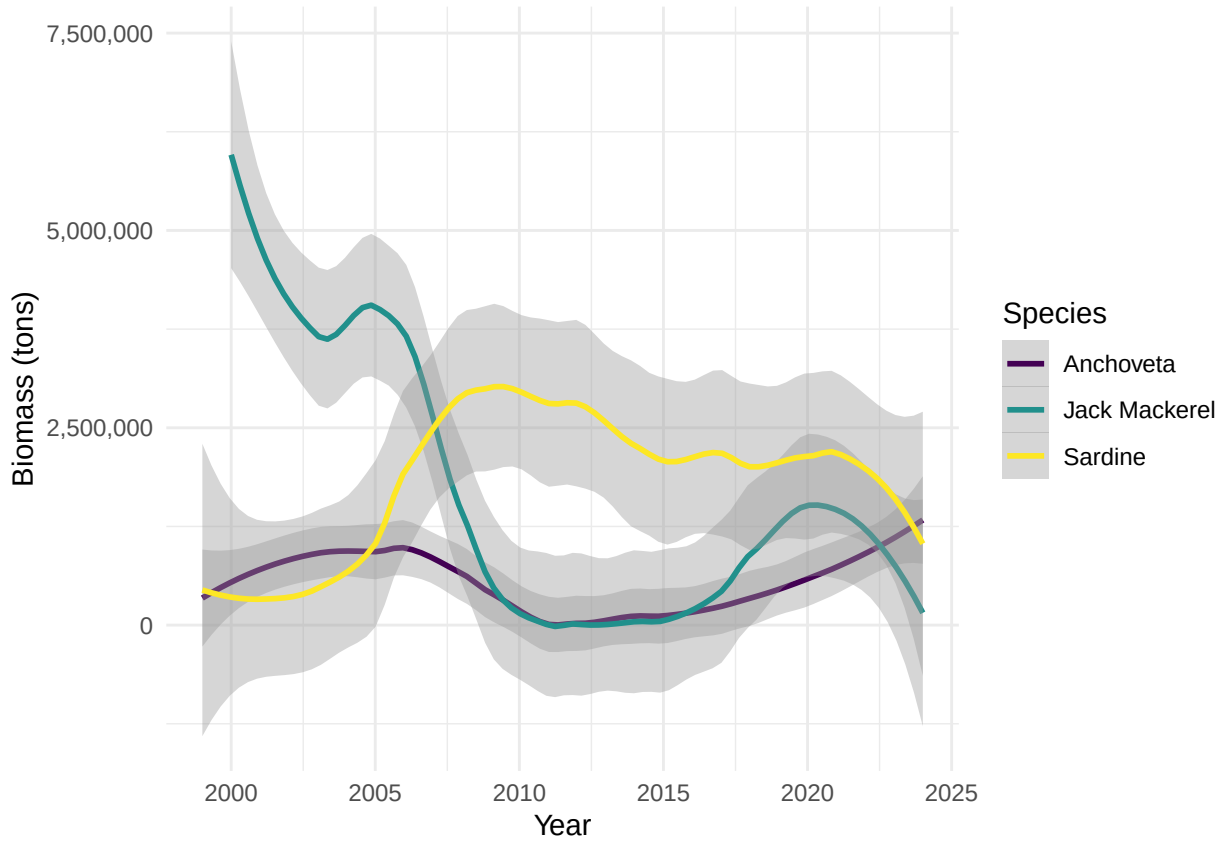

Figure 5: Estimated biomass of small pelagic species in Chile (2000–2024)

equation SUR system using the SEM framework implemented in lavaan, freely allowing the error terms across biomass equations to be correlated. The model is estimated via robust maximum likelihood (MLR), yielding SUR coefficients with Huber–White robust standard errors. Cross-species interaction terms were tested as an extension but did not improve model fit (see Section 4.1 and Appendix **??**).

3.3.2 Total annual trips

We model the annual number of fishing trips taken by vessel v in year y as a count process following Kasperski (**?**). Since the available logbook data only record purse-seine operations, the unit of observation is a vessel–year. Separate effort models are estimated for the industrial and artisanal fleets, allowing fishing activity to respond differently to economic conditions and regulatory constraints across sectors. This specification explicitly accommodates technological heterogeneity across fleet segments, reflecting differences in vessel capacity, operating scale, and production technology.

The baseline specification is a Poisson model:

$$T_{vy} \sim \text{Poisson}(\lambda_{vy}), \qquad \lambda_{vy} = \exp\big(U'_{vy}\beta\big)\,, \tag{4}$$

where T_{vy} denotes the total number of purse-seine trips recorded for vessel v in year y. Given substantial overdispersion in the data—the variance-to-mean ratio of T_{vy} is 22.4 for the artisanal fleet and 4.9 for the industrial fleet—we estimate a negative binomial (NB) model, which nests the Poisson as a special case. A likelihood ratio test strongly rejects the Poisson restriction in both fleets ($p < 0.001$).

The vector of explanatory variables U_{vy} includes output prices by species, vessel-level allocated harvest, fixed vessel characteristics, and operating conditions:

$$U_{vy} = [p_{sy},\, H^{alloc}_{vy},\, Z_v,\, Env_{vy}]. \tag{5}$$

Output prices p_{sy} correspond to species-specific ex-vessel prices paid by processing plants to fishers, obtained from IFOP's manufacturing survey and deflated to constant 2018 pesos using the consumer price index from the Central Bank of Chile. Consistent with Dresdner et al. (**?**), prices are measured as annual averages over peak fishing months to reflect the economically relevant conditions faced by vessels when planning annual effort.

Because the objective of the model is to characterize vessels' effort responses to changes in prices, quotas, and environmental conditions within the small pelagic fishery, we restrict attention to SPF trips recorded in logbooks. These trips account for approximately 95% of observed fishing revenue and therefore capture the primary economic margin through which vessels adjust annual fishing effort.

Quota shares do not enter Eq.~(**??**) directly. Instead, shares translate annual TACs into vessel-level allocated harvests. Let ω^r_{vs} denote vessel v's historical share of landings of species s within its administrative region r (for artisanal vessels) or regulatory zone (for industrial vessels), computed over the full sample period. Given the effective TAC $\bar{Q}^r_{sy}$ for species s in year y in region r—obtained from SERNAPESCA quota monitoring records and reflecting all inter-fleet transfers and adjustments—vessel-level allocated harvest is:

$$H^{alloc}_{vy,s} = \omega^r_{vs}\,\bar{Q}^r_{sy}, \qquad H^{alloc}_{vy} = \sum_s H^{alloc}_{vy,s}. \tag{6}$$

Each vessel is assigned to its administrative region based on its modal departure port from the logbook records. Artisanal TACs are assigned at the regional level (regions V, VIII, IX, XIV, X), while industrial TACs follow the regulatory zone structure established by SUBPESCA (zones V–IX and XIV–X for jack mackerel; zone V–X for sardine and anchoveta). This regional construction introduces cross-sectional variation in H^{alloc}_{vy} that reflects the heterogeneous quota environments faced by vessels operating in different parts of the Centro-Sur fishery.

The vector Z_v contains time-invariant vessel characteristics, including hold capacity (log of cubic meters) and vessel type. Hold capacity determines the physical constraint on catch per trip and proxies for vessel scale.

The vector Env_{vy} captures operating and regulatory constraints that vary across both vessels and years. Unlike the environmental variables in the stock dynamics model, which reflect biological productivity (SST, chlorophyll-a), the variables in the trip equation capture conditions that affect the *feasibility* of fishing operations. Two vessel–year variables are constructed using each vessel's center of gravity (COG)—the catch-weighted centroid of its haul locations over the sample period. First, the number of days with adverse weather conditions is computed as the count of days per year in which wind speed exceeds 8~m/s at the environmental grid point nearest to the vessel's COG.[2] Second, the number of days closed due to biological closures (*vedas*) is assigned based on the regulatory zone corresponding to the vessel's COG latitude. In the Centro-Sur region, sardine and anchoveta face seasonal closures for reproduction (August–October in regions V–VIII; July–October in regions IX–XIV) and recruitment (January–February in all regions), yielding 151 and 182 closed days per year, respectively. Jack mackerel has no biological closures.

Quota prices are not included explicitly in the trip equation. Instead, the economic scarcity of quota is captured implicitly through aggregate TAC levels and vessel-level allocated harvest.

Trips, harvest, and prices may be jointly determined within the year, implying potential endogeneity in reduced-form trip regressions. However, the purpose of Eq.~(**??**) is not causal identification but rather to provide an empirically grounded behavioral relationship that maps changes in prices, quotas, and environmental conditions into expected fishing effort under projected climate scenarios.

3.4 Projection approach

As noted by Kasperski (**?**), the multi-species bioeconomic framework is particularly suited for long-term projections of fish populations and fishing activity rather than intra-annual management responses. However, because our empirical models are estimated on interannual variability rather than decadal trends, the projections presented here capture the impact of changed environmental conditions without accounting for endogenous adaptation by fishers or managers (**?**). This provides a useful benchmark for understanding the direction and magnitude of climate impacts, while acknowledging that behavioral responses may attenuate or amplify these effects over time.

[2]The wind threshold of 8 m/s corresponds approximately to Beaufort scale 5, at which purse-seine operations become difficult for artisanal vessels. This threshold was selected based on model fit (AIC comparison across 8, 10, and 12 m/s thresholds); robustness checks are reported in Appendix **??**.

4 Results

4.1 Stock biomass model

Table **??** summarizes the model selection exercise. Five candidate specifications were compared, ranging from a baseline with only own-biomass terms to a full model including quadratic chlorophyll-a and cross-species interaction terms. The preferred specification—which includes own biomass, its square, SST, SST^2, and chlorophyll-a—yields the lowest AIC and BIC. Adding quadratic CHL terms or cross-species interactions does not improve fit; in fact, both AIC and BIC increase when these terms are included. Wind variables were excluded in an earlier specification search, as a joint exclusion test failed to reject their omission. The preferred specification contains six parameters per equation, yielding approximately 17 residual degrees of freedom with $N = 23$ observations.

Table **??** reports the SUR estimates of annual biomass dynamics for sardine, anchoveta, and jack mackerel. The system is estimated via robust maximum likelihood (MLR), yielding Huber–White standard errors and allowing unrestricted contemporaneous correlation across equations.

All three species exhibit strong interannual persistence and significant density dependence, consistent with discrete logistic growth. However, the species differ markedly in their sensitivity to environmental conditions.

Sardine biomass displays the strongest environmental response. Chlorophyll-a enters positively and is highly significant ($\hat{\rho}_{CHL} = 80.99$, $p < 0.001$), indicating that years with higher primary productivity favor sardine growth. The quadratic SST term is also significant and positive ($\hat{\rho}_{SST^2} = 54.74$, $p = 0.001$), suggesting a convex response to temperature anomalies—sardine biomass increases more sharply under larger SST deviations from the mean. This pattern is consistent with evidence from the Humboldt Current System, where sardine tends to exhibit smoother population changes associated with warmer or transitional oceanographic regimes (**?**; **?**).

Anchoveta biomass is dominated by density dependence, with the strongest own-biomass coefficient ($\hat{r} = 1.11$, $p < 0.001$) and a highly significant quadratic term ($\hat{\eta} = -0.202$, $p < 0.001$). SST has a marginally significant negative effect ($p = 0.087$), but neither SST^2 nor CHL are significant. This asymmetric environmental response—where anchoveta reacts primarily to temperature rather than productivity—is consistent with the ecological literature. Anchoveta in the Humboldt Current is known to respond sharply to short-term cold, nutrient-rich conditions, dominating during cold phases of decadal-scale oceanographic variability (**?**; **?**; **?**).

Jack mackerel shows strong persistence ($\hat{r} = 1.06$, $p < 0.001$) but a distinct environmental signature. The quadratic SST term is large and highly significant ($\hat{\rho}_{SST^2} = -56.49$, $p = 0.001$), indicating a clear thermal optimum: biomass declines when temperatures deviate substantially from the long-run mean in either direction. Neither linear SST nor CHL are individually significant, suggesting that jack mackerel growth is less sensitive to annual fluctuations in upwelling produc-

tivity than sardine. This weaker coupling to coastal environmental variability is consistent with jack mackerel's broader spatial distribution, deeper foraging behavior, and transboundary stock structure within the Southeast Pacific (**?**).

These contrasting environmental responses have direct implications for climate projections. Under warming scenarios, sardine may benefit from increased thermal variability if accompanied by sustained productivity, whereas anchoveta could face reduced biomass if warming erodes the cold, nutrient-rich conditions it depends on. Jack mackerel, with its thermal optimum response, would be adversely affected by both warming and cooling extremes.

Cross-species interactions were tested but did not improve model fit (Table **??**). The residual covariances across equations are also statistically insignificant, suggesting that shared unobserved shocks—beyond those captured by SST and CHL—do not generate strong contemporaneous co-movement across species. This does not rule out ecological interactions (e.g., predator–prey dynamics between jack mackerel and small pelagics), but it indicates that such effects are not detectable in the annual aggregate biomass data given the available sample size.

Appendix **??** presents three robustness checks. Table **??** restricts the jack mackerel series to observed (non-interpolated) years, reducing the sample to 12 observations and requiring a more parsimonious specification without SST^2. Table **??** estimates a two-species SUR for sardine and anchoveta only, treating jack mackerel biomass as an exogenous covariate. Table **??** augments the main specification with cross-species interaction terms, confirming that their coefficients are individually and jointly insignificant. The core findings—strong density dependence, positive CHL effects on sardine, and a thermal optimum for jack mackerel—are qualitatively robust across all specifications, lending confidence that the results are not driven by the interpolation procedure.

Table 3: Model selection for SUR biomass dynamics

Specification	K/eq	AIC	BIC	ΔAIC
No environment	3	490.7	507.7	13.1
Main (SST + SST2 + CHL)	6	477.6	504.9	0.0
Main + CHL2	7	482.9	513.6	5.3
Main + interactions	8	485.0	519.1	7.4
Full (interactions + CHL2)	10	490.5	528.0	12.9

4.2 Total Annual trips

Table **??** reports the negative binomial estimates separately for each fleet segment.

Allocated harvest enters positively and is highly significant in both fleets, confirming that vessels with larger quota allocations undertake more trips per year. This variable provides the key linkage between the trip equation and the simulation framework: changes in regional TACs translate into vessel-level effort adjustments through the quota share mechanism.

Table 4: SUR estimates of biomass dynamics (main specification, N = 23)

	Dependent variable:		
	Sardine	Anchoveta	Jack mackerel
	(1)	(2)	(3)
Own biomass (x_{it})	0.438^{***} (0.113)	1.112^{***} (0.157)	1.057^{***} (0.137)
Own biomass^2 (x_{it}^2)	-0.021^{***} (0.005)	-0.202^{***} (0.047)	-0.017^{***} (0.005)
SST	-9.158^{*} (5.409)	-4.972^{*} (2.906)	5.923 (5.161)
SST^2	54.738^{***} (16.038)	-4.608 (7.203)	-56.485^{***} (17.413)
Chlorophyll-a	80.992^{***} (17.061)	-5.366 (11.155)	15.454 (24.835)
Constant	23.745^{***} (2.132)	12.096^{***} (1.639)	28.188^{***} (3.376)
Observations	23	23	23
R^2	0.707	0.608	0.790
Adjusted R^2	0.621	0.493	0.729
Residual Std. Error (df = 17)	7.308	3.917	9.589

Note: *p<0.1; **p<0.05; ***p<0.01.
Robust (Huber–White) SEs via MLR. Biomass in 10^5 tons.
Environmental variables centered at sample means.
Jack mackerel uses Model B interpolation (see Section 3.1).

Table 5: Negative binomial estimates of annual fishing trips (2013–2024, $N_{IND} = 319$, $N_{ART} = 4498$)

	Dependent variable:	
	Industrial	Artisanal
	(1)	(2)
Hold capacity (log m^3)	-0.037 (0.201)	0.193^{***} (0.056)
Allocated harvest (H_{vy}^{alloc}, tons)	0.00003^{***} (0.00001)	0.001^{***} (0.00003)
Price jack mackerel (1000s \$/ton)	0.005^{***} (0.002)	-0.002^{**} (0.001)
Price sardine (1000s \$/ton)	-0.001 (0.002)	0.003^{***} (0.001)
Price anchoveta (1000s \$/ton)	-0.00001 (0.0003)	-0.0005^{***} (0.0002)
Bad weather days	-0.0001 (0.001)	-0.002^{***} (0.0004)
Veda days	-0.081^{***} (0.004)	0.009^{***} (0.002)
Vessel type (UNK)		-0.518^{***} (0.109)
Vessel type (BR)		-1.033^{***} (0.173)
Vessel type (BRV)		-2.456^{***} (0.189)
Vessel type (L)		-0.103 (0.150)
Vessel type (LM)	-0.982^{***} (0.077)	0.266^{*} (0.143)
Constant	14.638^{***} (1.995)	1.230^{***} (0.446)
Observations	319	4,201
Log Likelihood	$-1{,}231.493$	$-17{,}512.890$
θ	8.345^{***} (0.903)	1.407^{***} (0.031)
Akaike Inf. Crit.	2,480.986	35,051.770

Note: *p<0.1; **p<0.05; ***p<0.01.
Vessel-clustered robust SEs. Prices in constant 2018 thousands of pesos/ton.
Allocated harvest = historical vessel share × lagged aggregate harvest.
LR test NB vs Poisson: Industrial $\chi^2 = 524.8$; Artisanal $\chi^2 = 49060.9$ (both $p < 0.001$).

The two fleets differ markedly in their price responsiveness. For the industrial fleet, the jack mackerel price is positive and significant ($p < 0.01$), consistent with this fleet's primary orientation toward jack mackerel. Sardine and anchoveta prices have no detectable effect on industrial effort, reflecting the limited participation of industrial vessels in the sardine-anchoveta fishery. For the artisanal fleet, the sardine price is positive and significant ($p < 0.01$), consistent with sardine being the dominant target species for artisanal vessels in the Centro-Sur region. The anchoveta price enters with a negative and significant coefficient. This counterintuitive sign likely reflects simultaneity: years of low anchoveta availability produce both higher prices (through the inverse demand channel) and fewer trips (through reduced catch opportunities), generating a negative reduced-form correlation. As noted above, the purpose of this equation is not causal identification but rather to provide an empirically grounded mapping for the simulation, where prices are determined endogenously by the inverse demand module.

Hold capacity is not significant for the industrial fleet, where vessels are relatively homogeneous in scale, but is positive and significant for the artisanal fleet. Among artisanal vessels, larger hold capacity is associated with more annual trips, consistent with larger vessels being more commercially active and better able to sustain operations across varying conditions.

The environmental and regulatory variables show distinct patterns across fleets. For the artisanal fleet, adverse weather days are strongly negative and significant ($p < 0.001$), indicating that years with more days of wind speeds above 8~m/s substantially reduce fishing effort for smaller vessels. For the industrial fleet, adverse weather is not significant, consistent with larger vessels being better equipped to operate under rough conditions. Biological closure days are strongly negative for the industrial fleet, representing the largest effect in the model, consistent with the industrial fleet's dependence on access during open seasons. For the artisanal fleet, the positive coefficient on closure days reflects the current limitation that this variable varies only across regulatory zones (151 days for regions V–VIII versus 182 days for regions IX–XIV) and therefore captures locational differences in fishing activity rather than the causal effect of closures.[3]

Vessel type dummies capture residual heterogeneity in fleet composition. Among artisanal vessels, larger vessel categories (L, BRV) are associated with substantially fewer trips relative to the baseline, reflecting differences in trip duration and operational patterns across vessel classes.

4.3 Climate change projections

We assess the impact of climate change on fishing effort through two channels: (i) a *direct weather channel*, where projected changes in wind speed alter the number of bad weather days that constrain fishing operations, and (ii) an *indirect biomass channel*, where projected changes in sea surface temperature and chlorophyll-a affect stock growth capacity through the SUR equations, which in turn modifies harvest allocations and fishing effort through the trip equation.

[3]Once year-specific closure dates from annual SUBPESCA resolutions are incorporated, providing within-zone temporal variation, this coefficient is expected to become negative.